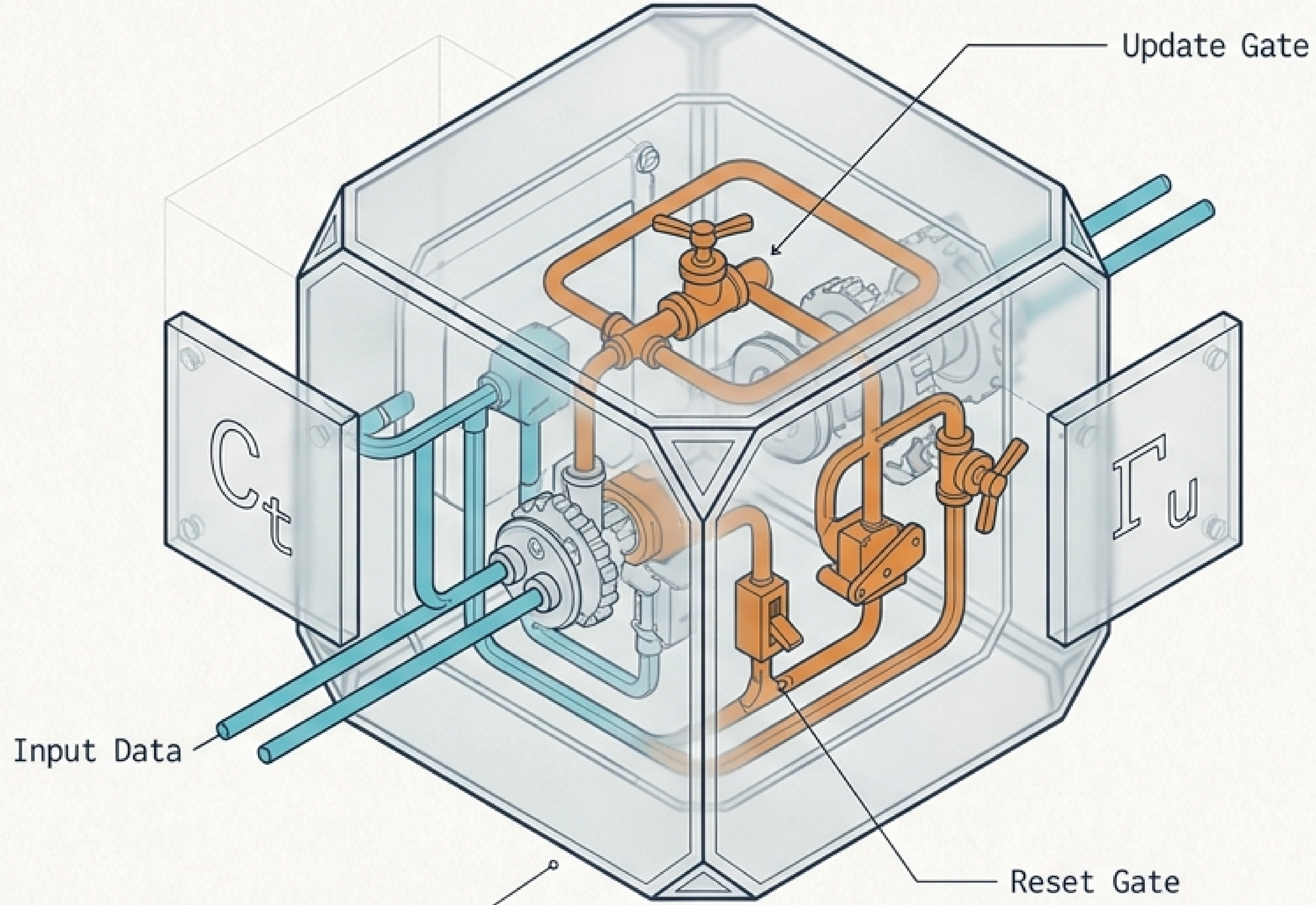
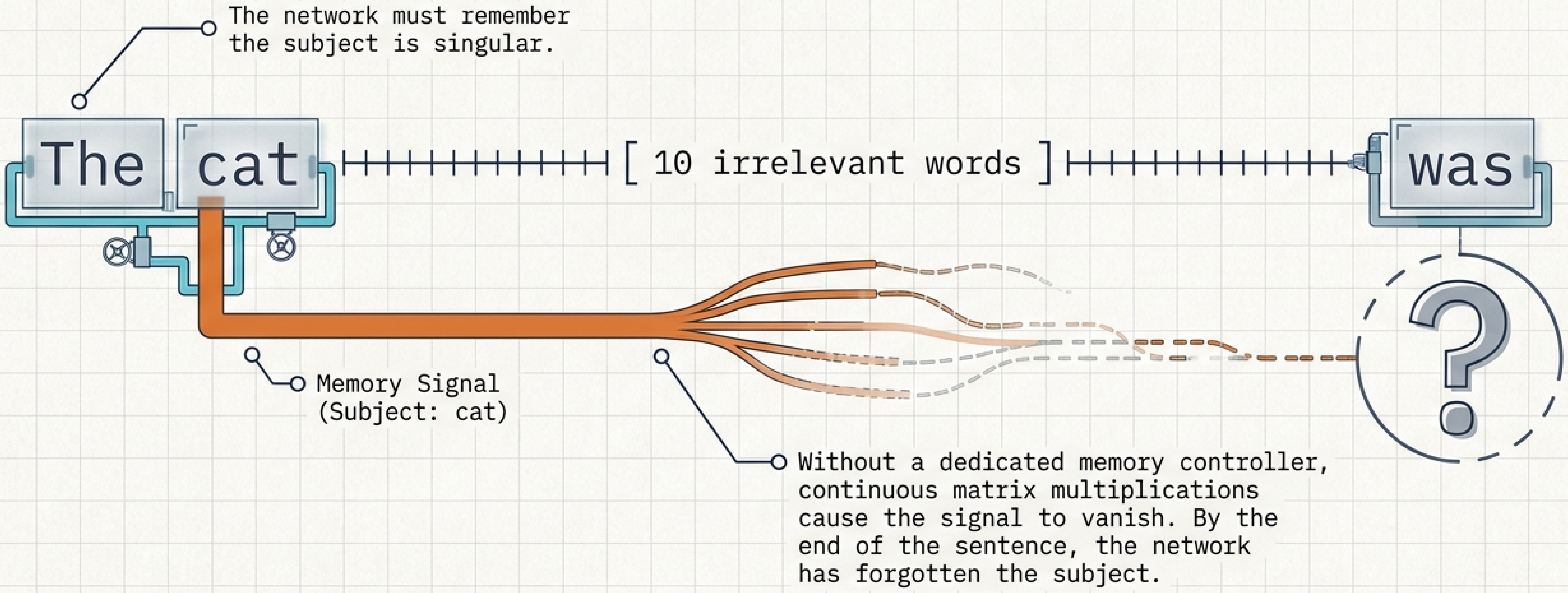
Decoding the GRU

The 80/20 blueprint for understanding Gated Recurrent Units.

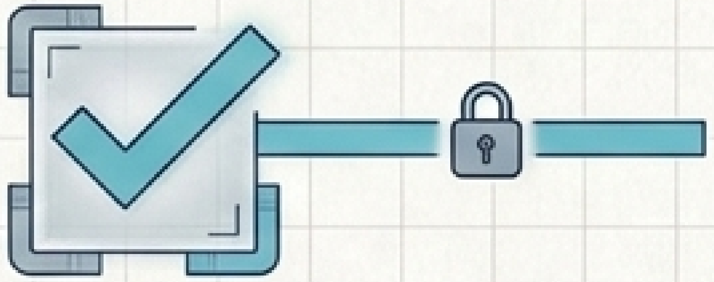
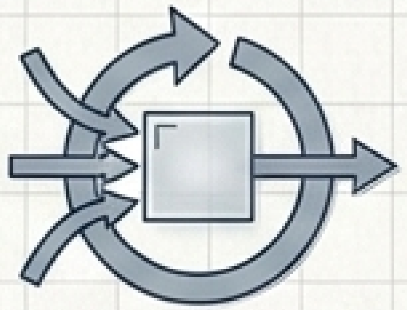
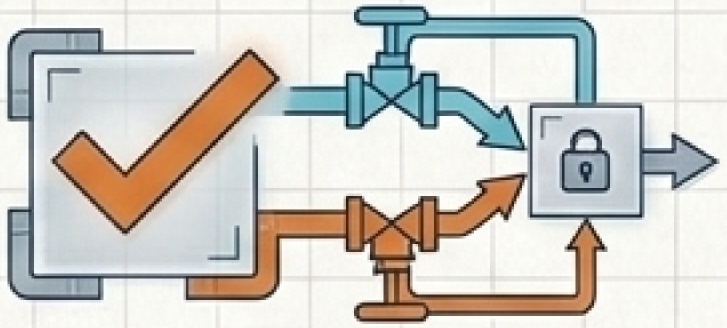
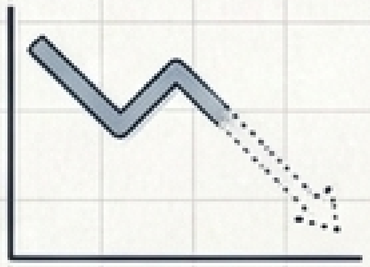
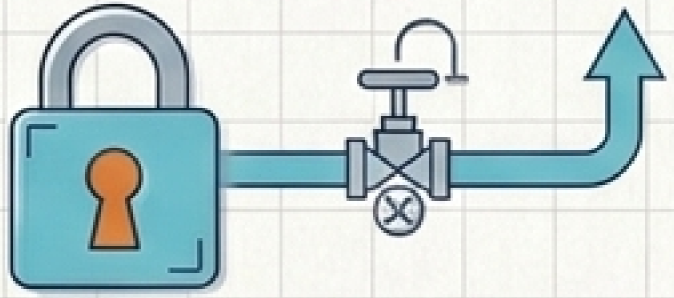


Architecture originally introduced by Chung, Gulcehre, Cho, & Bengio.

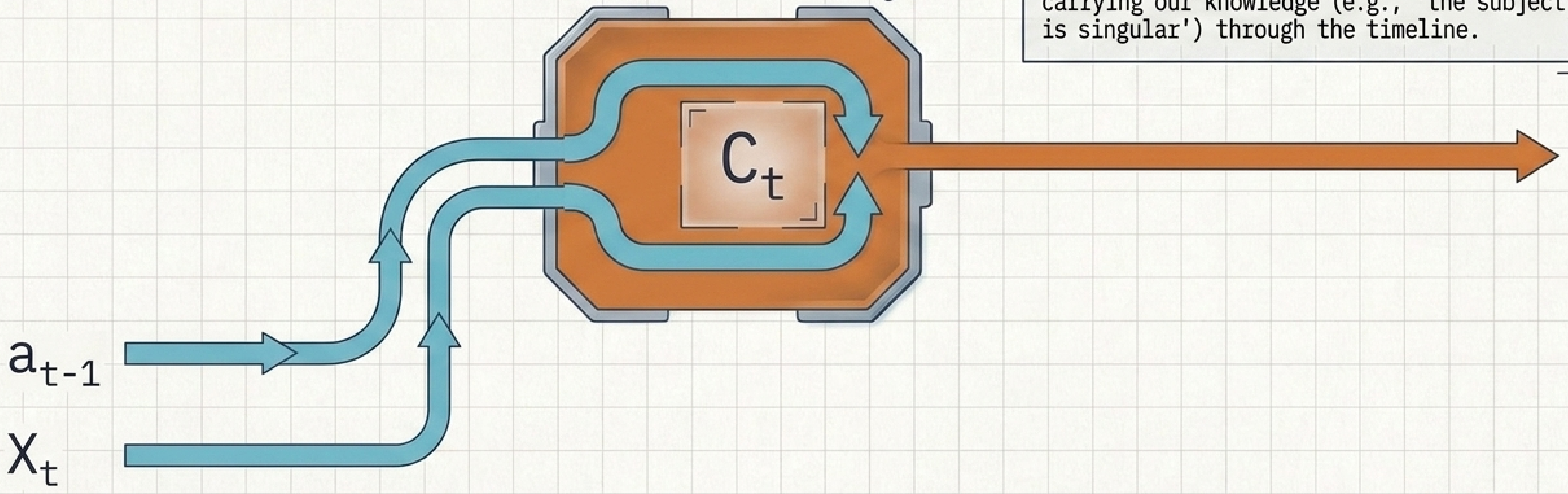
Standard RNNs suffer from catastrophic amnesia over long sequences.



The architectural leap from Standard RNN to GRU.

	Standard RNN		GRU	
Memory Span		Short-term (decays quickly)		Long-term (sustained across gaps)
Update Mechanism		Continuous Overwrite (every new word forces a change)		Gated Control (selective memorization)
Gradient Risk		Vanishing (signal dies)		Preserved (valves lock signal in place)

The GRU introduces a dedicated memory cell (C_t) to carry context forward.

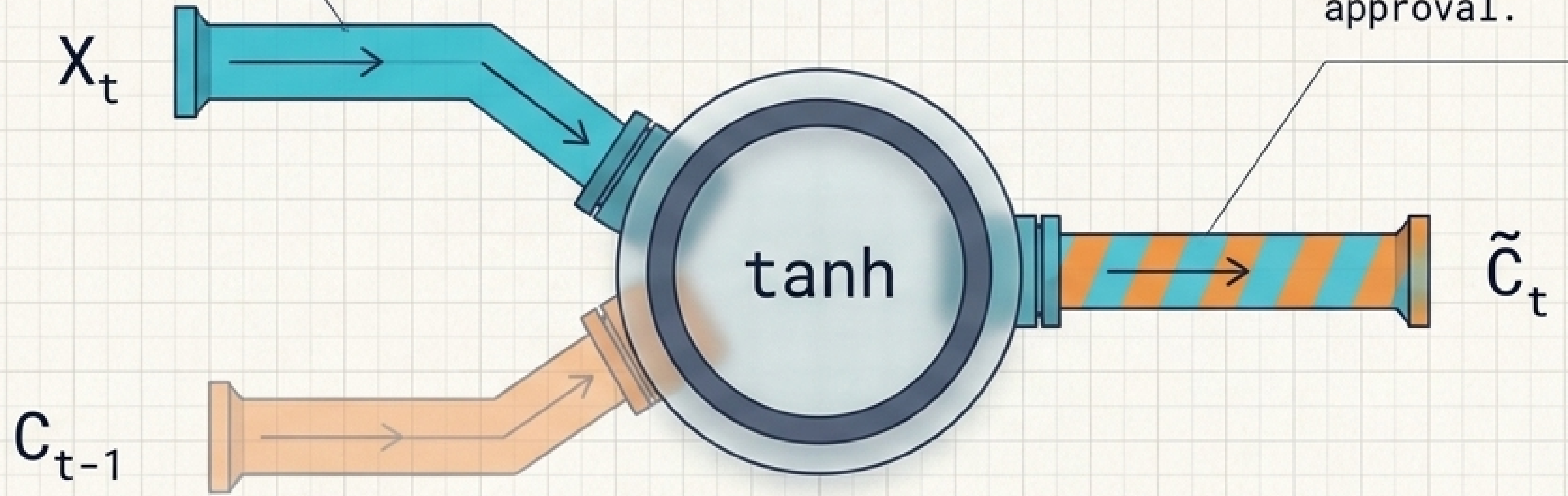


In a basic GRU, the memory cell value (C_t) is identical to the output activation (a_t). It acts as the physical vessel carrying our knowledge (e.g., 'the subject is singular') through the timeline.

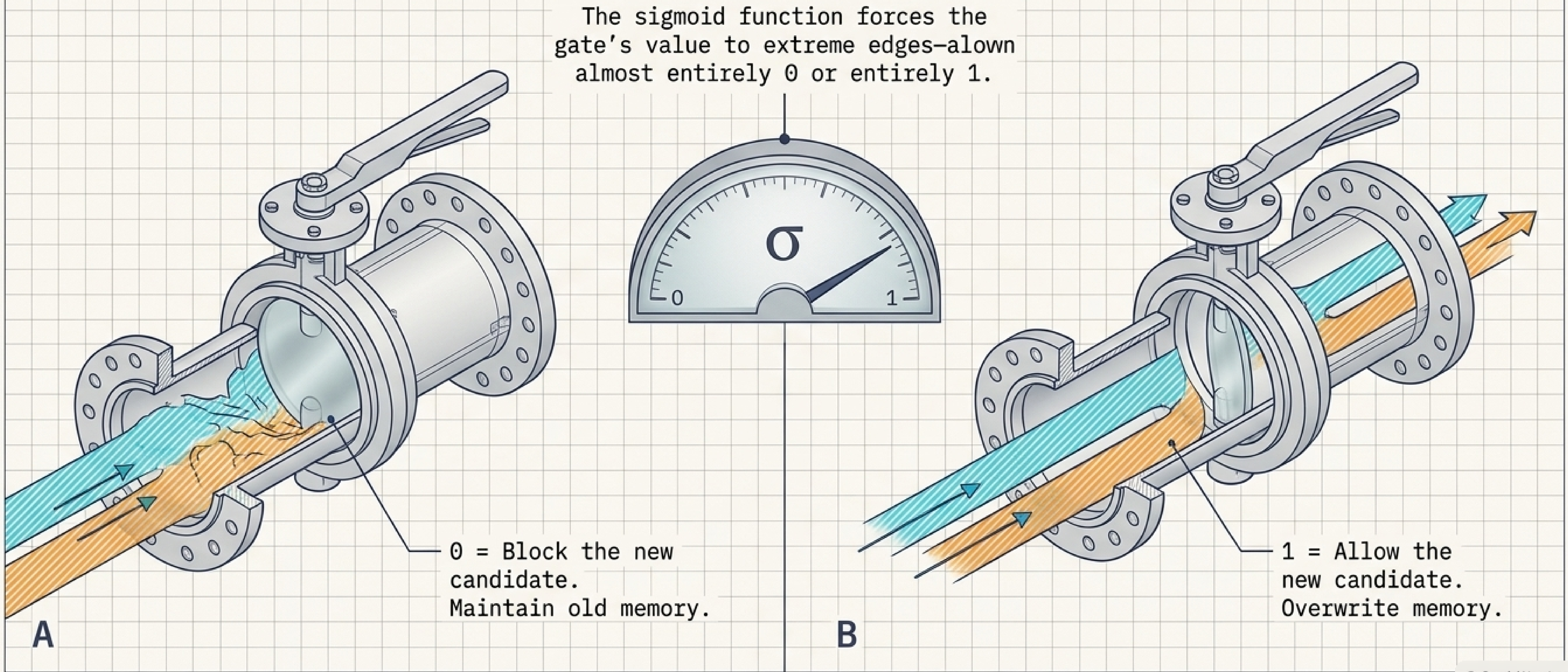
The Candidate ($\tilde{C}_t$) proposes a new memory based on current input.

The current word (X_t) and previous memory (C_{t-1}) are combined.

This creates a “draft” of a new memory. It is not permanent yet—it is only a candidate waiting for approval.



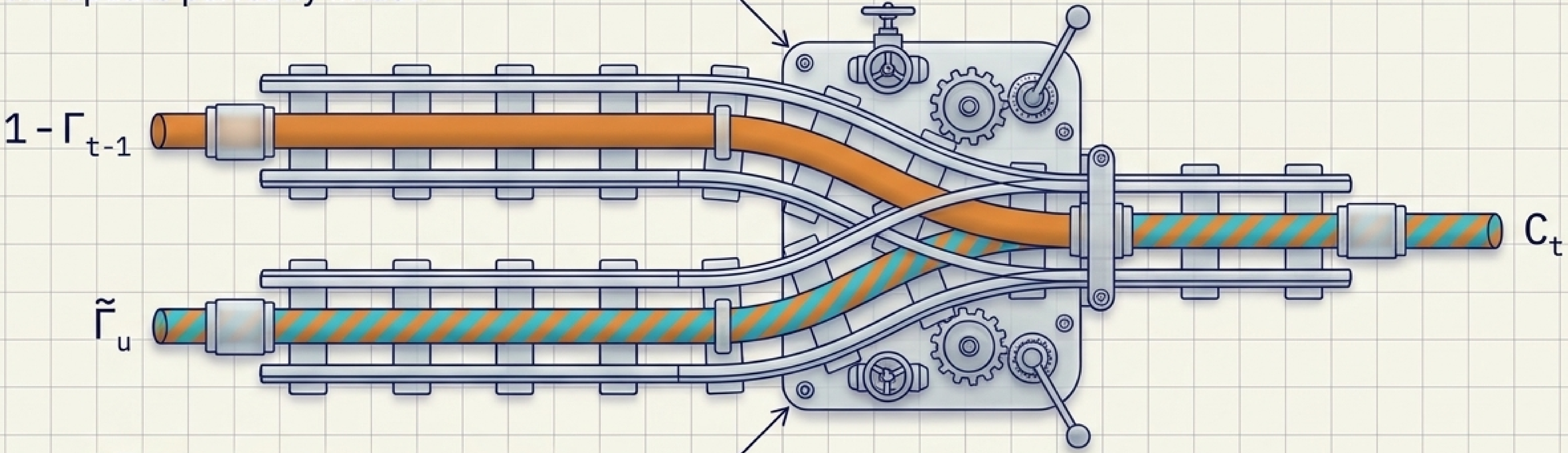
The **Update Gate** (Γ_u) acts as a mechanical valve guarding the memory.



$$C_t = \Gamma_u * \tilde{C}_t + (1 - \Gamma_u) * C_{t-1}$$

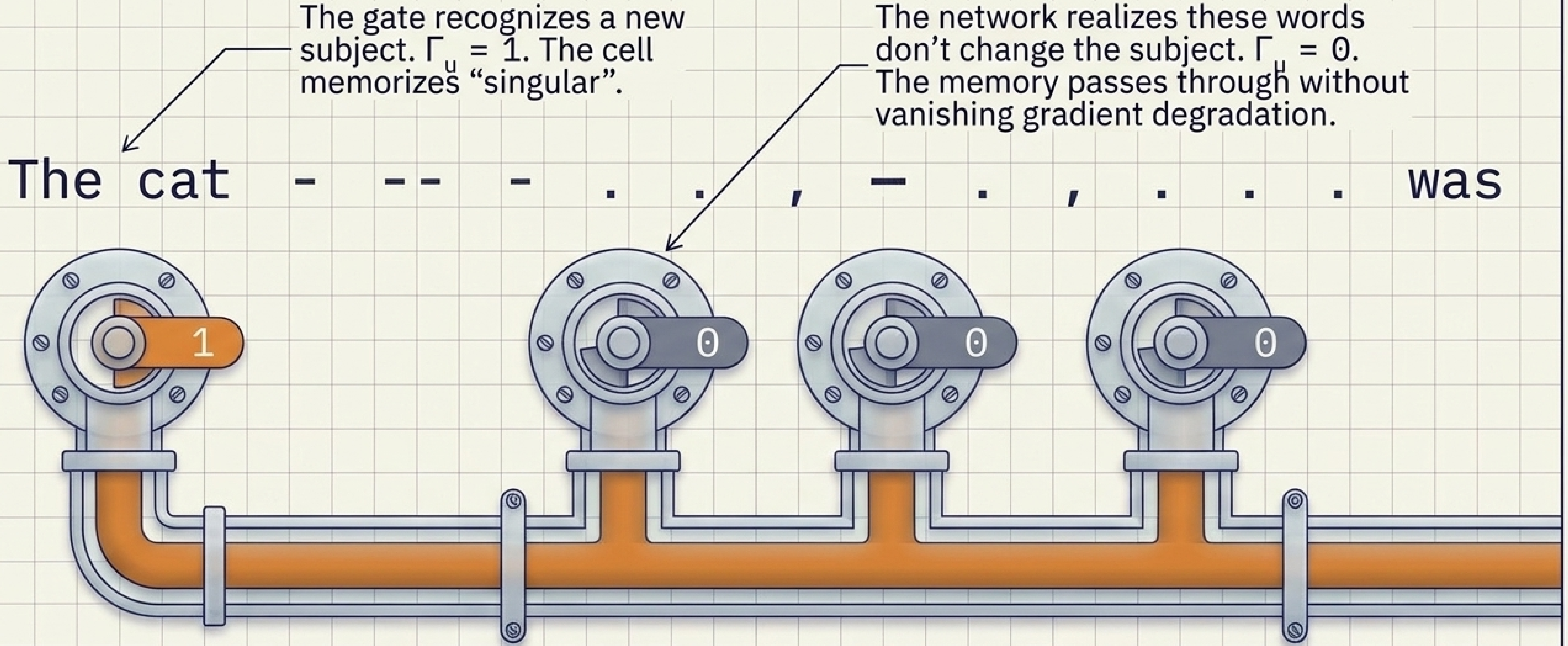
The core equation is a routing switch choosing between past and present.

If $\Gamma_u = 0$, the math cancels out the new idea. The old memory bypasses the update perfectly intact.

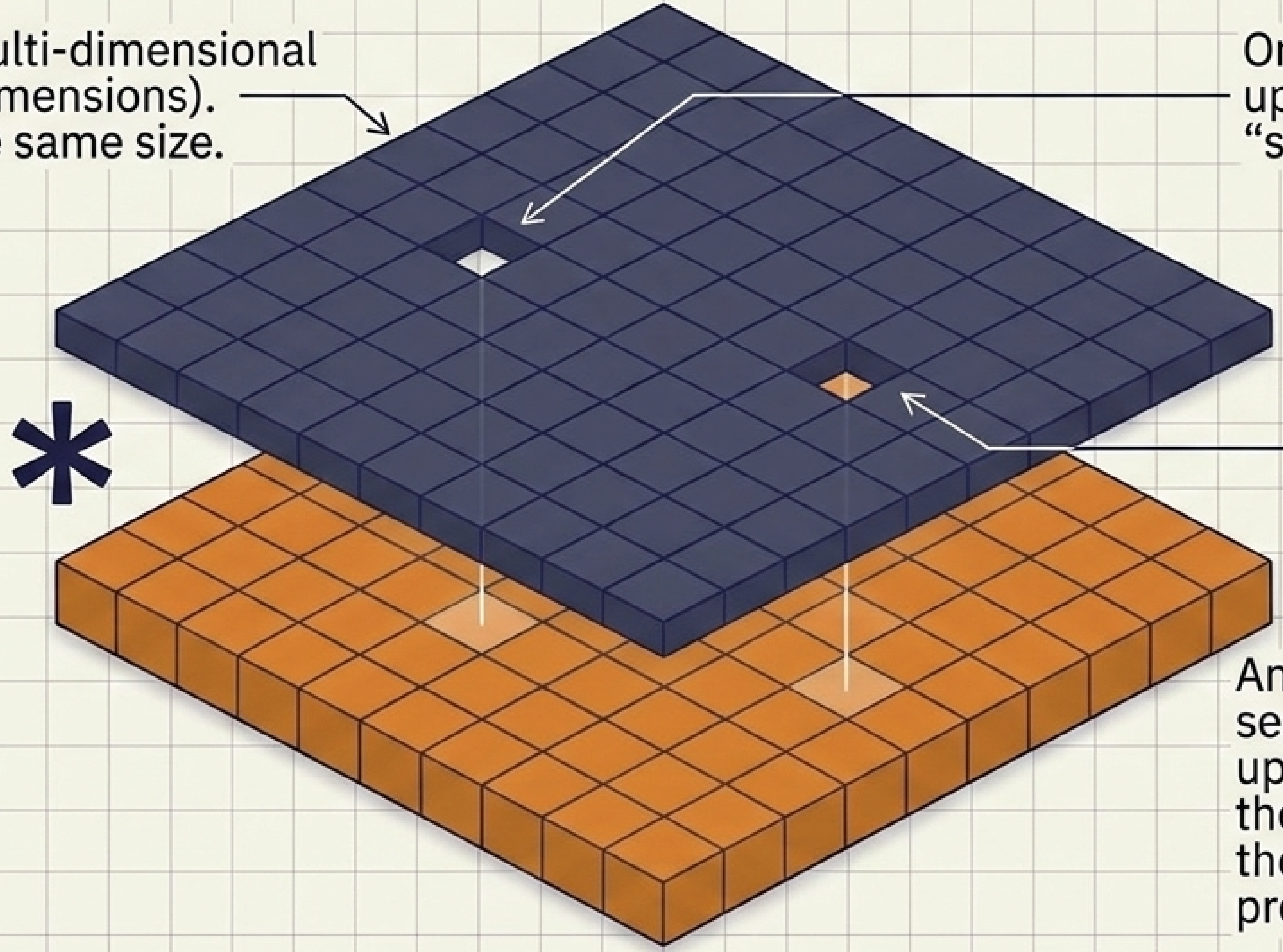


If $\Gamma_u = 1$, the math cancels out the old memory. The new candidate takes over entirely.

The **Update Gate** cures amnesia by locking the memory vault.



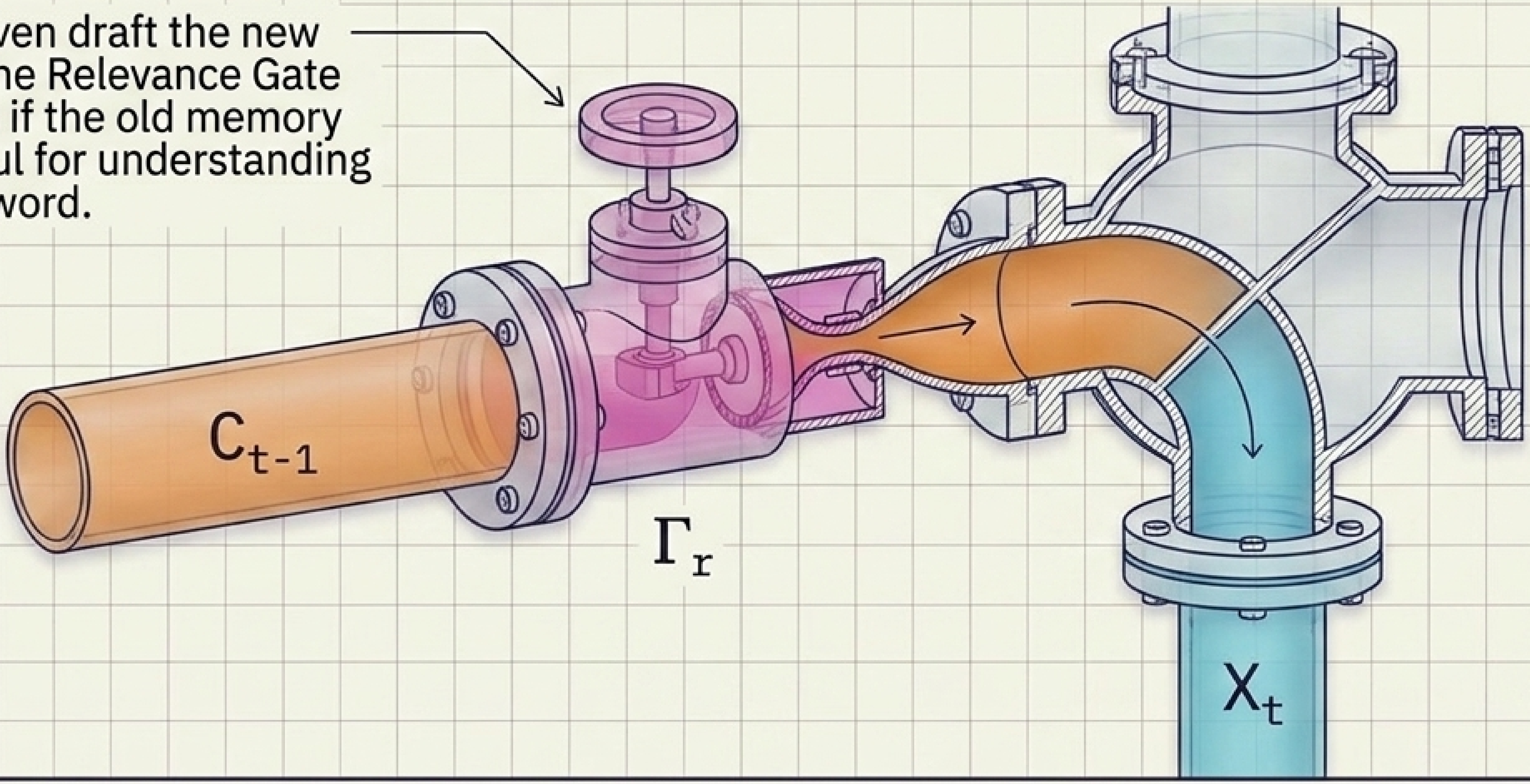
Element-wise gating acts as a stencil, allowing micro-targeted updates.

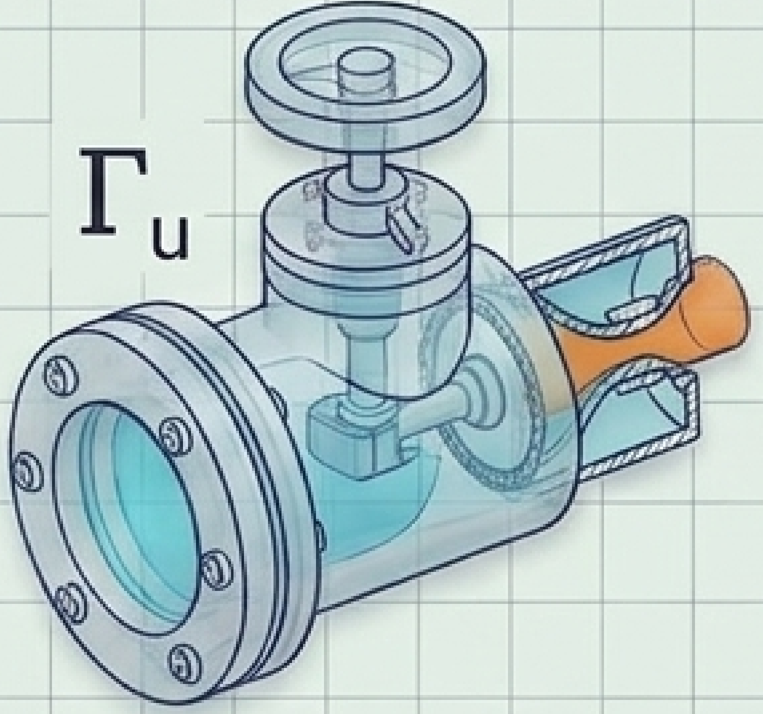


The full GRU architecture includes a second filter: The **Relevance Gate**.

$$\tilde{C}_t = \tanh(W_c[\Gamma_r * C_{t-1}, X_t] + b_c)$$

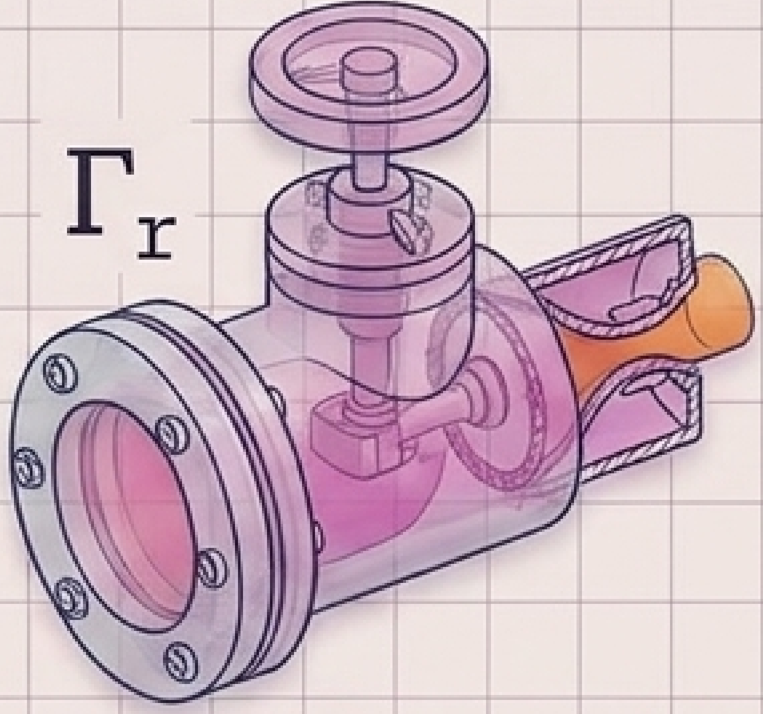
Before we even draft the new candidate, the Relevance Gate (Γ_r) decides if the old memory is even useful for understanding the current word.





Update Gate (Γ_u)

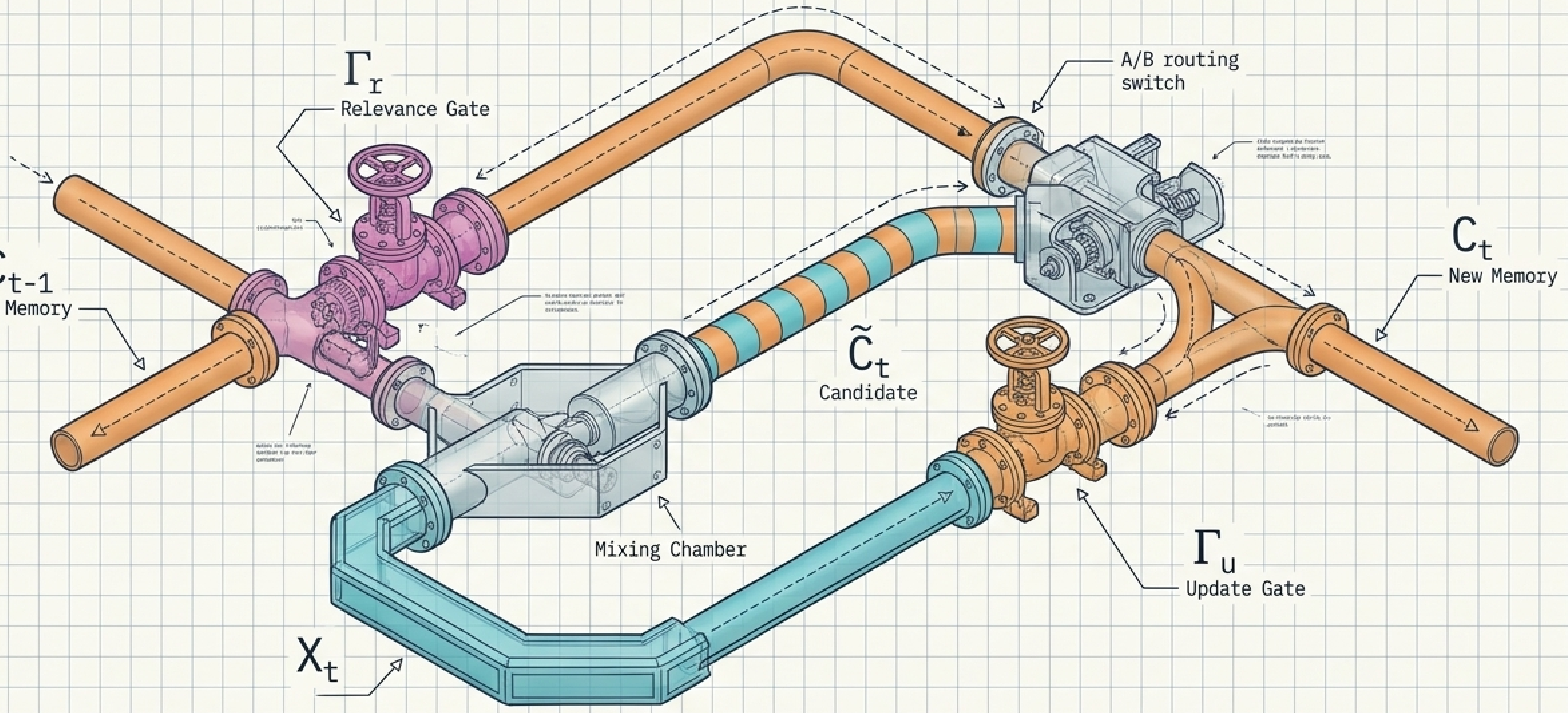
- **Question:** “Should I replace my long-term memory?”
- **Action:** Controls the final output equation. Protects the memory cell from vanishing gradients.



Relevance Gate (Γ_r)

- **Question:** “Does my old memory matter for this specific new word?”
- **Action:** Controls the candidate creation. Drops irrelevant past context when evaluating new inputs.

The **Gated Recurrent Unit:** A system of managed memory.



By transforming continuous matrix multiplication into a system of discrete, mechanical routing gates, the GRU solves the vanishing gradient problem, allowing neural networks to maintain context across vast sequences of data.